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Technical Reproducibility Validation Report

Ztest-Upload (Inventado → Slug Ztest-Upload-V0-7-Y

Version v0.7
Y-STR

Verification date	2026-07-19
Repository commit	b2033bfbb5cf55496b776463bdf2993fa763a4be
Attestation level	**Runs + Output Structure Validation**
Permalink	https://strhub.app/verified/ztest-upload-inventado-slug-ztest-upload-v0-7-y-v0-7-y
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution — confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

– Source available
– Installation successful
– End-to-end execution
– Output generated

NOT VERIFIED

– Genotype accuracy
– Concordance
– Forensic validity
– Regulatory compliance

1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	**5/5 gates passed — Runs + Output Structure Validation**
ATTESTATION LEVEL	**Runs + Output Structure Validation**
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. Reproducibility Metadata

Tool	Ztest-Upload (Inventado → Slug Ztest-Upload-V0-7-Y
Version	v0.7
Repository	https://github.com/tfwillems/HipSTR
Commit (pinned)	b2033bfbb5cf55496b776463bdf2993fa763a4be
Container OS	ubuntu-22.04
Marker panel	Y-STR
Verified on	2026-07-19 18:34:10 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/29698862349

3. Exact Run Command

The following command was executed verbatim in the CI environment. Replacing the input paths under /data/in with your own inputs reproduces the verified run.

```
# Environment: ubuntu-22.04 · Ztest-Upload (Inventado → Slug Ztest-Upload-V0-7-Y v0.7 · commit
b2033bf
$ HipSTR \
--bams \
/data/in/input.bam \
--fasta \
/data/ref/hg38.fa \
--regions \
/data/in/regions.bed \
--str-vcf \
/data/out/result.vcf.gz \
--min-reads \
5 \
--use-unpaired \
--read-qual-trim \
! \
--def-stutter-model \
--max-str-len \
250
```

4. Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS
Runs	Executes end-to-end without crashing	PASS
Expected IO	Produces a non-empty file in the declared format	PASS
Output Structure Validation	Output structure matches expected genotype-bearing format	PASS
Execution log	ztest-upload-inventado-slug-ztest-upload-v0-7-y-v0-7-y.log-external.txt	

5. Out of Scope

This report does not evaluate any of the following:

– Genotype correctness or accuracy
– Concordance against known truth sets
– Sensitivity, specificity, or stutter performance
– Allele calling accuracy or forensic casework suitability
– Regulatory compliance or ISO accreditation
– Multi-laboratory or multi-dataset reproducibility

6. Output Content Evidence

Output file	result.vcf.gz
Format	VCF
Sequence records	13
Distinct STR loci	13
Total reads	2,912
Max depth (single locus)	320

7. Verification Data

This tool does not include its own demo or test data. STRhub ran the verification using the public reference dataset listed below. That dataset is a slice around 14 forensic STR loci, not a whole genome: it carries reads only at the loci listed below.

Dataset	Illumina BAM (hg38) — HG002 (Y-STR, male)
Source	https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/dat…
License	Open access (GIAB / NIST). Research and educational use. STRhub is not the data provider.
Ref. genome	GRCh38 / hg38

Loci tested	DYS385_2 · DYS389I · DYS389II.1 · DYS390 · DYS391 · DYS392 · DYS393 · DYS438 · DYS456 · DYS458 · DYS635 · Y-GATA-A10 · Y-GATA-H4
Regions BED	Provided by the tool author — covers 14 of 14 supported loci

8. Verification Matrix

PASS	Reference dataset	Illumina BAM (hg38) — HG002 (Y-STR, male)
	README check (5/5)	Advisory — all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

9. Limitations

– Single reference dataset per input type
– Single containerized environment (Docker / ubuntu-22.04)
– No truth-set comparison or ground-truth genotypes
– No accuracy or concordance assessment
– No forensic validation of results
– Short-read limitations apply (very long STR alleles may not span reads)

10. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

11. Conclusion

◆ **Runs end-to-end**

Ztest-Upload (Inventado → Slug Ztest-Upload-V0-7-Y v0.7 installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit. All 5 verification gates were passed.

◆ **Produces valid output**

The tool generated a structurally valid VCF output file with genotype calls across 13 target forensic STR loci, with a maximum read depth of 320 at Y-GATA-H4.

◆ **No bundled demo data**

Ztest-Upload (Inventado → Slug Ztest-Upload-V0-7-Y does not include its own test or demo dataset. STRhub Verified supplied Illumina BAM (hg38) — HG002 (Y-STR, male) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

◆ **Reproducibility statement**

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.

Appendix — Detected Markers with Read Depth

Ztest-Upload (Inventado → Slug Ztest-Upload-V0-7-Y v0.7 · verified 2026-07-19

LOCUS	READ DEPTH		LOCUS	READ DEPTH	
Y-GATA-H4	320		**DYS456**	196	
DYS389II.1	298		**DYS391**	195	
DYS389I	272		**Y-GATA-A10**	194	
DYS635	258		**DYS458**	193	
DYS385_2	219		**DYS392**	190	
DYS390	218		**DYS393**	161	
DYS438	198				